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QiandaoEar22: a high-quality noise dataset for identifying specific ship from multiple underwater acoustic targets using ship-radiated noise

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Abstract

Target identification of ship-radiated noise is a crucial area in underwater target recognition. However, there is currently a lack of multi-target ship datasets that accurately represent real-world underwater acoustic conditions. To tackle this issue, we conducted experimental data acquisition, resulting in the release of QiandaoEar22—a comprehensive underwater acoustic multi-target dataset. This dataset encompasses 9 h and 28 min of real-world ship-radiated noise data and 21 h and 58 min of background noise data. To demonstrate the availability of QiandaoEar22, we executed two experimental tasks. The first task focuses on assessing the presence of ship-radiated noise, while the second task involves identifying specific ships within the recognized targets in the multi-ship mixed data. In the latter task, we extracted eight features from the data and employed six deep learning networks for classification, aiming to evaluate and compare the performance of various features and networks. The experimental results reveal that ship-radiated noise can be identified from background noise in over 99% of cases. For the specific identification of individual ships, the optimal recognition accuracy achieves 99.56%. Finally, we found using spectrum and MFCC as feature inputs and DenseNet as classifier can achieve excellent recognition performance. Considering the computational efficiency, CRNN and ECAPA-TDNN are also good choices. Our work not only establishes a benchmark for algorithm evaluation but also inspires the development of innovative methods to enhance underwater acoustic target detection (UATD) and underwater acoustic target recognition (UATR).

Keywords: Underwater acoustic dataset, Underwater target recognition, Underwater acoustics, Deep learning

1 Introduction

The significance of underwater target sound as a vital source of information has been underscored since the invention of sonar by American Fessenden in 1913 [1]. Consequently, underwater acoustic technology has been progressively applied, facilitating the differentiation of various sounding target types and enabling analysis of environmental

signals. Notably, the focus has shifted toward the ship-radiated noise due to its enhanced concealment and security.

However, publicly available datasets for underwater acoustic target research are scarce for two primary reasons. Firstly, military secrecy restricts access to numerous recordings made in military facilities [2]. Secondly, the high cost associated with conducting underwater experiments, compounded by signal loss during propagation and limited data quality due to underwater communication equipment constraints, presents challenges in dataset creation [3, 4]. Data labeling further contributes to the time and energy consumption.

Some small-scale datasets have been created, including those from cargo ships, navy vessels, and personal watercraft [2, 5–9]. While some recent datasets such as ShipsEar [10] and Deepship [11] are available, they primarily consist of single-target ship data in simple environments, lacking the complexity of multi-target scenarios often encountered in real-world conditions.

To address this issue, we present the QiandaoEar22 dataset, containing 21 h and 58 min of background noise data and 9 h and 28 min of multi-target ship noise signal data. Collected using digitalHyd SR-1 self-capacitance hydrophones in a real underwater acoustic environment at Qiandao Lake, China, during June 2022, this dataset serves as a high-quality ship-radiated noise recording.

Underwater acoustic target identification focuses on discerning the class of ship-radiated noise from background noise. Additionally, underwater acoustic target detection aims to determine the presence or absence of ship targets through binary classification, classifying received signals as either ship-radiated or background signals. This preliminary classification streamlines subsequent tasks. To accomplish this, we first identify ship-radiated noise and background noise signals from acquired underwater acoustic data. Extracting three 1D features (spectrum, power spectral density (PSD), and detection of envelope modulation on noise (DEMON) spectra [12]) and five 2D features (low-frequency analysis and recording (LOFAR) spectrum [13], Log Mel spectrum, Mel-frequency cepstral coefficients (MFCC) [14], power-normalized cepstral coefficients (PNCC) [15], and Gammatone frequency cepstral coefficients (GFCC) [16]) for the multi-target ship sound signal. Then, we employ diverse deep neural networks, including convolutional neural network (CNN) [17], convolutional recurrent neural network (CRNN) [18], bidirectional long short-term memory (BiLSTM) [19], densely connected convolutional network (DenseNet) [20], residual network (ResNet) [21], and emphasized channel attention, propagation, and aggregation in time delay neural network (ECAPA-TDNN) [22] to classify these features. Finally, three types of ship signals: SpeedBoat, KaiYuan, and underwater unmanned vehicle (UUV) are identified from the multi-target ship signals, respectively, as shown in Fig. 1. Furthermore, we explore the labeling and formatting aspects of the dataset. Our objective is that the dataset, along with the experimental results, assists researchers in benchmarking algorithms and promotes the development of innovative methods to improve UATD and UATR systems.

The main contribution of this paper is as follows. An underwater acoustic multi-target dataset named QiandaoEar22, is constructed through the experiment conducted in Qiandao Lake in 2022. Then, two experimental tasks are demonstrated and to recognize the ship-radiated noise from background noise and to identify the specific ship from the



Fig. 1 Target ships

multiple targets. During the second experimental task, we compare the performance of different features and network, which could be references when using this dataset for other potential purposes.

Section 2 details the experiments on acquisition and labeling of the QiandaoEar22 dataset. Section 3 introduces the main methodology of feature extraction, classification, and the indicators. Section 4 introduces the evaluation of experimental tasks, encompassing the experimental setup, presentation of the two tasks, and analysis and discussion of the results. Conclusions are presented in Sect. 5.

2 The QiandaoEar22 dataset

2.1 Recording system and methodology

The data collection took place from June 24 to 28, 2022, at the Xin'anjiang Experimental Site of the Institute of Acoustics, Chinese Academy of Sciences (CAS), situated in the town of Qiandao Lake, Zhejiang, China. Throughout the deployment, a hydrophone was anchored to the center of the deployment area using a counterweight and buoy system, as depicted in Fig. 2. The water depth ranged from approximately 30 to 50 ms, and the hydrophone was positioned at a depth of 10–15 ms. A dedicated hydrophone was employed for the data collection, using a stand-alone setup positioned near the channel to capture a broader range of radiated noise from the ships.

The hydrophone model used is DigitalHyd SR-1, illustrated in Fig. 3, with key specifications provided in Table 1. The DigitalHyd SR-1 hydrophone has a sample frequency of 52,734 Hz, a sample resolution of 16 bits, a usable acoustic band of 125.8 kHz, a receive sensitivity of $-162.2 \sim -126.1$ dB re 1V/ μ Pa, and an input sound pressure level range of 46.3 dB re 1 μ Pa $\sim$ 172.5 dB re 1 μ Pa. The operating depth can reach up to 100 ms, constructed with plastic steel, weighing approximately 0.77 kg. It operates within a temperature range of 0–40 °C and employs a USB interface. The device is powered by a rechargeable lithium-ion battery, enabling continuous data collection for up to 12 h. Acoustic data are stored in WAV format on a removable memory card, which also retains all configuration parameters for subsequent data analysis.

2.2 Dataset structure

During the hydrophone deployment, we observed ships navigating through the channel. We recorded the time when a ship approached and moved away from the hydrophone, approximately 1 km away, along with the ship number and name at those moments. Subsequent data alignment involves identifying the waveform peak and associating it with the corresponding ship number, thereby correlating the labeling

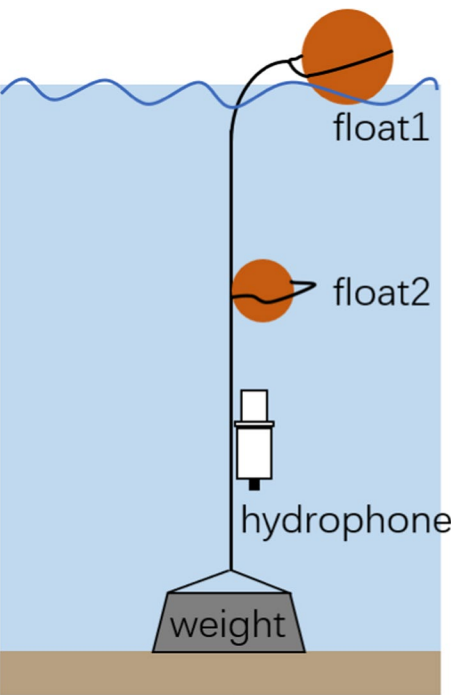


Fig. 2 Schematic diagram of the experimental hydrophone to be deployed in Qiandao Lake in June 2022



Fig. 3 DigitalHyd SR-1 hydrophone

Table 1 DigitalHyd SR-1 hydrophone specifications

Index	Parameter
Sample frequency	52,734 Hz
Sample resolution	16 bits
Usable acoustic band	1–25.8 kHz
Receive sensitivity	−162.2 ~ −126.1dB re 1V/μPa
Input sound pressure level range	46.3 dB re 1μPa–172.5 dB re 1μPa
Operation depth	100 m
Surface material	Plastic steel
Weight	0.77 kg
Operation temperature	0–40 °C
Device interface	USB interface
Power supply	Rechargeable lithium-ion battery

and data for that specific segment. The data collection cite is a tourist attraction, resulting in recorded ship-radiated noise primarily originating from sightseeing boats and SpeedBoats. Due to the hydrophone’s proximity to the navigation channel,

passing vessels, including sightseeing boats, SpeedBoats, and other watercraft, contribute to a composite radiated noise. Additional targets include large tour boats, cargo ships, sanitation vessels, small boats, research ships, and ambient noise.

Based on recorded information, signal amplitude, and spectral characteristics, we determined the start and end time nodes for each audio segment. We labeled each data based on auditory sensation, distance, and their types. During this process, the data were segmented into 3-s intervals to simplify the subsequent processing. The final database consists of 10,611 records in WAV format, totaling 9 h, 28 min, and 13 s of ship-radiated noise data. Each audio file, along with its corresponding data, forms a record in the QiandaoEar22 dataset. The example information fields for each record are as following:

20220624102345 _ DATA0008 _ M _ GreenCity _ N _ S & SpeedBoat _ M _ W
(start/stop time _ single/multiple targets(S/M) _ ship name _ distance(N/M/F) _
audibility(S/M/W) & name _ distance _ audibility)

- “filename”: filename;
- “time”: start and end time of the audio recording;
- “single/multiple target”: the audio capture object is a single ship target or multiple ship targets, which is denoted by S for single target and M for multiple targets;
- “name”: name of the vessel;
- “distance”: indicates the distance of the vessel from the hydrophone, N for near, M for medium, and F for far;
- “audibility”: indicates the audibility of the voice, S for strong, M for Middle, and W for weak;
- If the audio includes multiple targets, link each target with ‘&’ in order, and also include the “name,” “distance,” and “audibility” information tag for each ship.

Among these tags, the auditory sensation is a subjective indicator which means the sound intensity of what the human ears hear, which can be strong, middle, or weak. Based on the consistency of labels, we identified a total of 143 recordings containing both single and multiple ship targets. These data primarily include 20 categories of ship targets, and the overall time distribution of data containing individual targets is illustrated in Fig. 4.

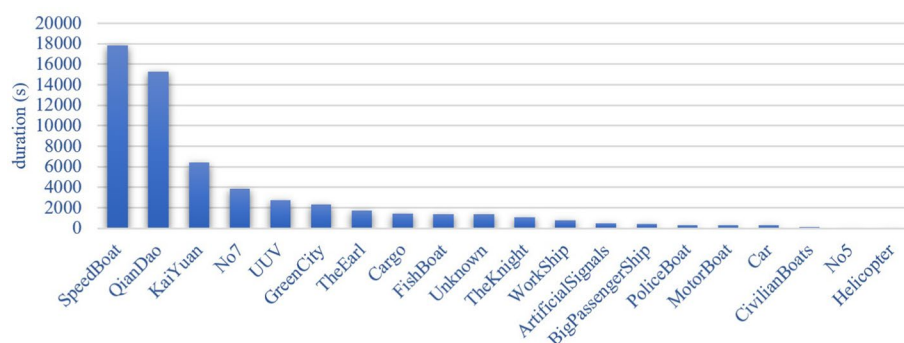


Fig. 4 Duration of the recorded data for different targets

Simultaneously, we selected time slices without targets and other interference to constitute the background noise data, labeled as files DATA0XXX. Ultimately, we obtained a total of 21 h and 58 min, comprising 25,900 records of background noise data, each with a duration of 3 s. The dataset can be download following the instructions on <https://github.com/xiaoyangdu22/QiandaoEar22>.

3 Methodology

3.1 Feature extraction

In this paper, we extract three types of one-dimensional features from the multi-target ship sound signals, including the spectrum, PSD, and DEMON spectrum, as well as five types of two-dimensional features, including the LOFAR spectrum, Log Mel spectrum, MFCC, PNCC, and GFCC. A brief introduction to these feature extraction methods is provided below.

3.1.1 Spectrum

The spectrum, also known as the frequency spectrum density, is obtained by applying the Fourier transform to convert the signal from the time domain to the frequency domain. We extract the single-sided spectrum, which is represented by the following formula:

$$F(j\omega) = \sum_{n=0}^{N-1} x(n)e^{-j\omega n}, 0 \leq \omega \leq \pi \quad (1)$$

N is the length of the signal sequence. Then, we take the absolute value of the above expression, which gives us the desired amplitude spectrum.

$$\text{Amplitude}(A(\omega)) = |F(j\omega)| \quad (2)$$

3.1.2 PSD

The power spectrum reflects the variation of signal power with frequency within a unit frequency band. In this paper, the Welch method is used to estimate the power spectral density. First, the data $x, t = 0, 1, \dots, T - 1$ of length N is divided into L segments, each with M data points. The data in the i -th segment are represented as follows:

$$x_i(t) = x(t + iM - M), 0 \leq t \leq M, 1 \leq i \leq L \quad (3)$$

Then, the window function $w(t)$ is applied to each data segment, and the periodogram of each segment is computed. The periodogram of the i -th segment is given by:

$$I_i(\omega) = \frac{1}{U} \left| \sum_{t=0}^{M-1} x_i(t)w(t)e^{-j\omega t} \right|^2, i = 1, 2, \dots, M - 1 \quad (4)$$

U is referred to as the normalization factor,

$$U = \frac{1}{M} \sum_{t=0}^{M-1} w^2(t) \quad (5)$$

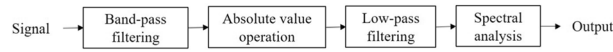


Fig. 5 DEMON spectral feature extraction process

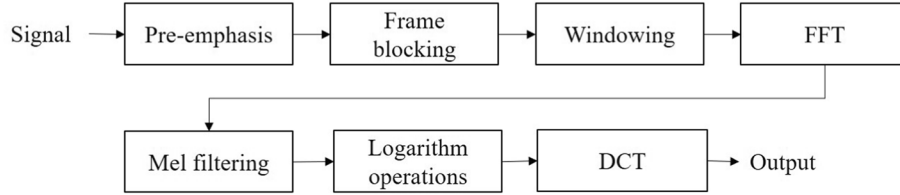


Fig. 6 MFCC feature extraction process

By assuming that the periodograms of each segment are approximately uncorrelated, the final expression for the power spectrum estimation is obtained as follows:

$$P_{xx}(e^{j\omega}) = \frac{1}{L} \sum_{i=1}^{I_d} I_i(\omega) \quad (6)$$

3.1.3 DEMON spectra

For underwater acoustic targets such as ships, propeller noise is the primary source of noise. The DEMON spectral analysis method demodulates the high-frequency broadband continuous spectrum noise to obtain the envelope information of the propeller spectral lines. The flowchart of this method is shown in Fig. 5. The demodulation of different frequency bands results in varying spectral line intensity characteristics. Referring to [12], we selected signals within the 1–10,000 Hz range to plot the DEMON spectra.

3.1.4 LOFAR spectra

LOFAR spectrogram is essentially a low-frequency line spectrogram. The steps to obtain the LOFAR spectrogram are as follows: Firstly, we normalize and centralize the signal. Then, we perform short-time fourier transform (STFT) on the signal and plot the corresponding time-frequency spectrogram,

$$STFT(t, \omega) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)e^{-i\omega\tau}d\tau \quad (7)$$

where $x(t)$ denotes the time-varying signal to be analyzed, and $w(t)$ is the time window function.

3.1.5 MFCC

MFCC, short for Mel-frequency cepstral coefficients, is an acoustic feature that aligns with the auditory mechanism of the human ear. The steps for MFCC feature extraction including pre-emphasis, framing, windowing, FFT, Mel filtering, LOG operations, discrete cosine transform (DCT), which are illustrated in Fig. 6.

3.1.6 Log Mel spectra

In the field of audio signal processing, in addition to the widely used MFCC method, the Log-Mel spectra is also a popular feature extraction technique. While both are based on the Mel frequency scale, they differ in their processing approaches. MFCC extracts low-dimensional features by applying a DCT to the logarithmic Mel spectrum, whereas the Log Mel spectra directly performs a logarithmic transformation on the Mel spectrum to generate easily processable spectral features. The expression for Mel frequency is as follows:

$$Mel(f) = 2595 \times \lg \left(1 + \frac{f}{700} \right) \quad (9)$$

In the formula, f represents the linear frequency. Since the human ear's perception of sound is not linear, we calculate the logarithmic energy of each filter bank output for easier processing. The Log Mel spectra retains more frequency information, making it particularly effective in certain audio classification and recognition tasks.

3.1.7 GFCC

Gammatone filter is a filter bank based on the auditory model of the human cochlea, and it can effectively simulate the frequency-selective characteristics of the basilar membrane. Its time-domain expression is as follows:

$$g_i(t) = At^{n-1}e^{-2\pi b_i t} \cos(2\pi f_i + \varphi_i)v(t), t \geq 0, 1 \leq i \leq N \quad (10)$$

In the formula, A is the filter gain, n is the filter order, N is the number of filters, f is the center frequency, φ is the phase, and b is the attenuation factor. GFCC is similar to MFCC. After converting the pre-processed audio signal into the frequency domain, the energy spectrum is passed through the Gammatone filter bank, followed by logarithmic compression and discrete cosine transform to obtain the GFCC.

3.1.8 PNCC

Power-normalized cepstral coefficients (PNCC) were introduced by Kim in 2016 [15]. Compared to MFCC, PNCC enhances the robustness of speech recognition while maintaining recognition performance. PNCC starts by filtering the audio signal and decomposing it into different frequency bands using a gammatone filter bank. It then calculates the energy spectrum for each band, followed by a series of nonlinear processing steps, such as logarithmic compression and power normalization. These steps are intended to reduce the impact of background noise on feature extraction.

3.2 Deep learning classification algorithms

After feature extraction, this paper employs deep neural networks, including CNN, CRNN, BiLSTM, DenseNet, ResNet, and ECAPA-TDNN, to classify the features.

3.2.1 CNN

Convolutional neural networks (CNN), as one of the most popular and widely applied models in the field of deep learning in recent years, are primarily composed of convolutional layers, pooling layers, and fully connected layers. The convolutional layers contain varying numbers of convolutional kernels depending on the network's requirements, and these parameters are continuously updated during backpropagation to achieve optimal performance. By stacking multiple convolutional layers, the network learns deep features of the data, followed by a rectified linear unit (ReLU) layer. Finally, in the pooling layer, the feature matrix obtained from the previous layer is compressed through average pooling or max pooling operations, resulting in a smaller matrix that meets the task's requirements.

3.2.2 CRNN

CRNN [18] is a network that combines CNN with recurrent neural networks (RNN). Since RNNs suffer from the vanishing gradient problem, making it difficult to capture extensive contextual information, we use LSTM instead. The special design of LSTM allows it to capture long-distance dependencies. By using this network, the contextual relationships within the audio can be learned, effectively improving recognition accuracy and making the model more robust.

3.2.3 BiLSTM

Traditional neural networks typically propagate and train data from front to back during training. This training method does not fully utilize the inherent information contained within sequences, resulting in low data utilization. The BiLSTM network is composed of a forward LSTM layer and a backward LSTM layer. This structure allows the network to utilize both past and future states to improve prediction accuracy, addressing the deficiency of insufficient data information in standard LSTM networks.

3.2.4 DenseNet

As shown in Fig. 7, DenseNet is fundamentally composed of dense blocks, transition layers, convolutional layers, and fully connected layers. It reuses feature maps through dense connections, leveraging feature maps from different layers to reduce inter-layer dependency. By providing compact and diversified input features through shortcut connections of varying lengths, DenseNet effectively mitigates the gradient vanishing problem that is difficult to optimize in deep networks. The final prediction is made using features from all layers, enabling better performance and model robustness on standard datasets with a smaller model size and reduced computational workload.



Fig. 7 Structure of DenseNet

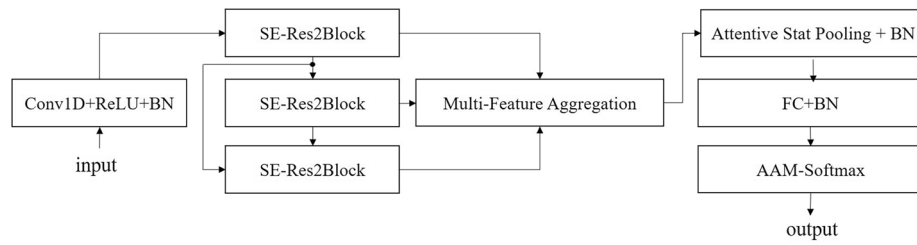


Fig. 8 Structure of ECAPA-TDNN

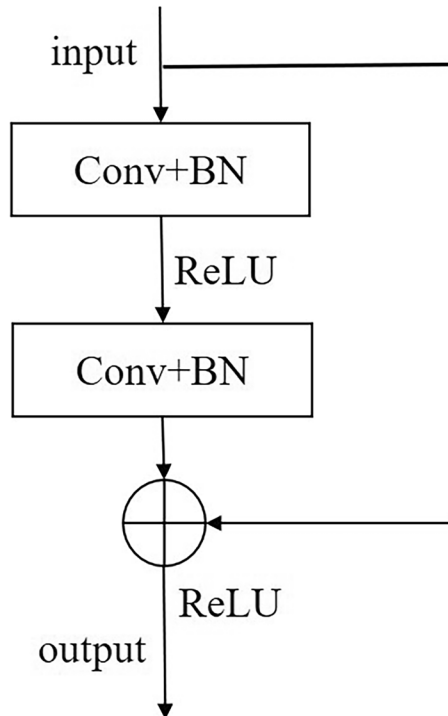


Fig. 9 Structure of residual block

3.2.5 ECAPA-TDNN

ECAPA-TDNN is a type of time delay neural network (TDNN), which includes squeeze-and-excitation (SE) blocks, residual networks, and a temporal self-attention mechanism. Its structure is shown in Fig. 8. ECAPA-TDNN applies statistical pooling to project variable-length signals into fixed-length target feature embeddings. Specifically, ECAPA-TDNN employs a SE-RES2Block with channel attention functionality, effectively enhancing the feature extraction performance of the neural network. Additionally, ECAPA-TDNN utilizes a multi-layer feature aggregation structure and densely connected network to generate statistical features.

3.2.6 ResNet

The main idea of ResNet is to use skip connections between the input and output channels to enhance the connections between different channels, thereby preventing the loss

of information. In this paper, the ResNet18 network is employed, which primarily consists of four stacked residual blocks. The structure of a residual block is shown in Fig. 9. It connects the input feature information to the output of the residual block through a 1×1 convolution, alleviating the issues of vanishing and exploding gradients that can occur during the training of deep networks.

3.3 Indicators

In this paper, recall, precision, accuracy, F1 score, false-negative rate (FNR), and false-positive rate (FPR) are used to evaluate the model's performance.

1. Accuracy: the proportion of correctly classified samples. The formula for accuracy is as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (11)$$

2. Recall: the proportion of positive examples that are correctly predicted. The formula is as follows:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (12)$$

3. Precision: the proportion of predicted positive examples that are actually correct. The formula is as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (13)$$

4. F1 score: a new metric that considers both precision and recall, which is defined as follows:

$$F1\text{-score} = 2 * \frac{\text{Precision} * \text{TPR}}{\text{Precision} + \text{recall}} \quad (14)$$

5. FNR: the proportion of missed positive samples to the total number of actual positive samples is as follows:

$$FNR = \frac{FN}{FN + TP} \quad (15)$$

6. FPR: the proportion of false-positive samples to the total number of actual negative samples is as follows:

$$FPR = \frac{FP}{FP + TN} \quad (16)$$

Here, true positive (TP) refers to the number of positive samples correctly predicted as positive, false negative (FN) is the number of positive samples incorrectly predicted as negative, true negative (TN) denotes the number of negative samples correctly predicted as negative, and false positive (FP) represents the number of negative samples incorrectly predicted as positive.

4 Ship noise detection and classification and analysis of results

Underwater acoustic target detection focuses on discerning the presence or absence of a target and classifying the received signal as either underwater target radiated noise or non-target noise, involving binary classification. Hence, using this dataset, we undertook two tasks. Task 1 involves separating ship signals from background noise in the whole acoustic recording. In Task 2, our objective is to identify three types of ship signals—SpeedBoats, KaiYuan, and UUVs—from the mixed multi-target ship-radiated noise signals.

4.1 Task 1: determine whether the audio contains ship-radiated noise

The objective of Task 1 is to determine whether the audio contains ship-radiated noise. We ultimately selected the first 5000 ship noise recordings, along with the initial 30 recordings from DATA0001 and 20 recordings from DATA0002, to compose the final dataset. Data segmentation details have been introduced in Part 2. We use 70% of the dataset for training, 15% for validation, and 15% for test. The characteristics of the final dataset are presented in Table 2. Log Mel spectra features were extracted for the data. Deep learning networks were employed for feature classification, and the network structures used are shown in Table 3. Each experiment was conducted ten times to calculate the mean and variance.

Table 4 presents the accuracy results for all deep learning methods. Except BiLSTM, the classification accuracy for other methods exceeded 99%. Notably, the CRNN exhibited the most robust classification performance, achieving an accuracy of 99.79%. Additionally, its recall, precision, F1 score, FNR, and FPR values also reached 99.79%, 99.80%, 99.79%, 0.00%, and 0.43%, respectively. The experimental results indicate a high level of distinguishability between ship-radiated noise data and background noise data within the dataset.

4.2 Task 2: determine if there are any targets of interest in the multi-ship mixed data

In Task 2, our primary goal is to determine the presence of any targets of interest in the multi-ship mixed data. Due to the data collection site is a tourist attraction, the collected ship-radiated noise mainly comes from passenger ships, SpeedBoats, and similar vessels. When selecting target ships, we want them belong to different categories and have sufficient data for subsequent experiments. So, the selected research objects include SpeedBoats, passenger ship KaiYuan, and UUV. Corresponding datasets were constructed for each research object, as illustrated in Table 5. Subsequently, examples of the extracted features and their corresponding dimensions are depicted in Fig. 10.

Table 2 Background target ship dataset

Dataset	Train		Test		Valid		Total	
Kind	Target	Back	TArget	Back	Target	Back	Target	Back
Num	3500	3550	750	700	750	750	5000	5000
Duration (s)	10,500	10,650	2250	2100	2250	2250	15,000	15,000
Total time (s)	21,150		2350		4500		30,000	

Table 3 The structure of deep learning network

Network	Structure
CNN	Conv ¹ (cin = 1, cout = 4, k = 5, s = 2, p = 1)+ BN(4) + Dropout(0.1)+ ReLU+ MaxPooling(k = 2, s = 2) Conv(cin=4, cout = 16, k = 5, s = 2, p = 1)+ BN(16) + Dropout(0.1)+ ReLU+ MaxPooling(k = 2, s = 2) Conv(cin = 16, cout = 32, k = 3, s = 2, p = 1)+ BN(32) + Dropout(0.1)+ ReLU+ MaxPooling(k = 1, s = 1) Conv(cin = 32, cout = 64, k = 3, s = 2, p = 1)+ BN(64) + Dropout(0.1)+ ReLU+ MaxPooling(k = 1, s = 1) Linear(64*128, 4*128)+ Linear(4*128, 32)+ Linear(32, 2)
CRNN	Conv(cin=1, cout=4, k = 5, s = 2, p = 1)+ BN(4) + Dropout(0.1)+ ReLU+ MaxPooling(k = 2, s = 2) Conv(cin = 4, cout = 16, k = 5, s = 2, p = 1)+ BN(16) + Dropout(0.1)+ ReLU+ MaxPooling(k = 2, s = 2) Conv(cin = 16, cout = 32, k = 3, s = 2, p = 1)+ BN(32) + Dropout(0.1)+ ReLU+ MaxPooling(k = 1, s = 1) Conv(cin = 32, cout = 64, k = 3, s = 2, p = 1)+ BN(64) + Dropout(0.1)+ ReLU+ MaxPooling(k = 1, s = 1) LSTM(256, 512, 2) Linear(512, 32) + Linear(32, 2)
BiLSTM	LSTM(input, 256, 2)
ResNet	ResNet18 [21](num classes = 2)
DenseNet	DenseNet121 [20]
ECAPA-TDNN	ECAPA-TDNN [22]

¹ Note: Conv1d is used for 1D features and Conv2d for 2D features

Table 4 Results (%) of Task 1

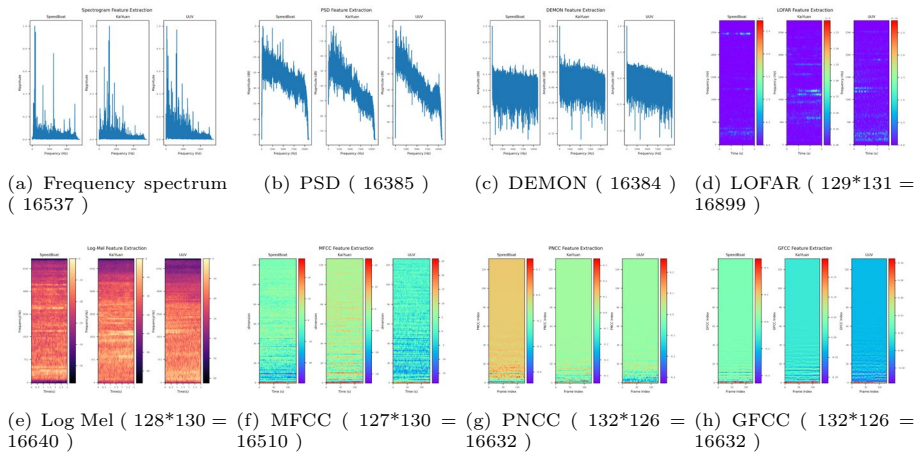
Net	Accuracy	Recall	Precision	F1	FNR	FPR
CNN2d	99.66±0.00	99.65±0.00	99.66±0.00	99.65±0.00	0.27±0.00	0.43±0.00
CRNN2d	99.79±0.05	99.79±0.09	99.80±0.18	99.79±0.05	0.00±0.13	0.43±0.00
DenseNet	99.72±0.04	99.72±0.06	99.73±0.06	99.72±0.05	0.13±0.21	0.43±0.06
ECAPA-TDNN	99.31±0.00	99.42±0.00	99.29±0.01	99.42±0.00	0.67±0.02	0.71±0.01
BiLSTM	91.17±0.01	91.37±0.00	91.61±0.01	91.17±0.01	14.27±0.03	0.30±0.00
ResNet	99.52±0.01	99.50±0.01	99.53±0.01	99.52±0.01	0.13±0.03	0.86±0.01

To facilitate performance comparison among different features, we constrained the feature dimensions within the range of 16,384 (128*128) to 16,900 (130*130). Finally, deep learning networks were employed for feature classification, with the network structures consistent with Task 1. For 1D features, CNN, CRNN, and BiLSTM were exclusively used for comparative experiments. Model performance was evaluated using recall, precision, accuracy, F1 score, FNR, and FPR.

For the three target datasets established, we conducted various comparative experiments with different features and networks. To better analyze the experimental results, we summarize the results from different perspectives, forming the following three sections.

Table 5 The SpeedBoat, KaiYuan, and UUV datasets

Data		Train		Test		Valid	
SpeedBoat	Category	SpeedBoat	Other	SpeedBoat	Other	SpeedBoat	Other
	Num	4848	2578	1041	554	1038	552
	Duration (s)	14,544	7734	3123	1662	3114	1656
	Total time (s)	22,278		4785		4770	
KaiYuan	Category	KaiYuan	Other	KaiYuan	Other	KaiYuan	Other
	Num	1363	2801	293	599	291	600
	Duration (s)	4089	8403	879	1797	873	1800
	Total time (s)	12,492		2676		2673	
UUV	Category	UUV	Other	UUV	Other	UUV	Other
	Num	631	1401	136	299	135	300
	Duration (s)	1893	4203	408	897	405	900
	Total time (s)	6096		1305		1305	

**Fig. 10** The examples of feature extraction. To facilitate performance comparison among different features, we constrained the feature dimensions within the range of 16,384 (128*128) to 16,900 (130*130)

4.2.1 Comparison of network

In this section, to compare the performance of different networks, we calculated the average indicators for 1D and 2D features separately in experiments with different datasets using the same network. The results in the figures represent all UUV experiment outcomes with varying shades of blue bars, all KaiYuan experiment results with varying shades of green bars, and similarly, all SpeedBoat experiment outcomes with varying shades of orange bars.

Comparison of network for 2D feature classification

The experimental results of different networks for 2D feature classification are shown in Table 6 and Fig. 11. We can observe that in the UUV experiment, the network performance from best to worst is DenseNet, ResNet, CRNN, CNN, ECAPA-TDNN, and BiLSTM. In the KaiYuan experiment, the network performance from best to worst is DenseNet, CRNN, ResNet, CNN, ECAPA-TDNN, and BiLSTM. In the SpeedBoat experiment, the network performance from best to worst is DenseNet, ECAPA-TDNN,

Table 6 Average results (%) of deep learning methods for 2D feature classification. To compare the classification performance of different networks, we averaged the experimental results of all 2D features under the same network and dataset

Experiment	Net	Accuracy	Recall	Precision	F1 score	FNR	FPR
UUV	DenseNet	97.78	96.96	97.93	97.36	4.34	0.84
UUV	ResNet	97.56	96.75	97.59	97.08	5.43	1.07
UUV	CRNN2D	96.93	95.45	97.45	96.30	8.45	0.59
UUV	CNN2D	96.21	94.81	96.35	95.48	8.93	1.46
UUV	ECAPA-TDNN	96.15	94.99	95.97	95.41	8.10	1.65
KaiYuan	DenseNet	94.52	93.24	94.39	93.69	10.51	3.02
KaiYuan	CRNN2D	92.73	91.17	92.32	91.60	13.36	4.29
SpeedBoat	DenseNet	92.61	91.82	91.92	91.83	5.60	10.76
KaiYuan	ResNet	92.49	90.63	92.17	91.30	14.77	3.96
KaiYuan	CNN2D	92.19	90.33	91.93	90.98	15.10	4.24
SpeedBoat	ECAPA-TDNN	92.08	90.78	91.65	91.14	4.97	13.48
KaiYuan	ECAPA-TDNN	91.82	90.00	91.31	90.45	15.30	4.70
SpeedBoat	ResNet	90.85	89.43	90.33	89.78	5.90	15.25
SpeedBoat	CNN2D	89.14	87.98	88.21	88.00	8.23	15.81
SpeedBoat	CRNN2D	89.05	88.22	88.00	87.94	9.11	14.22
UUV	BiLSTM	88.48	84.68	87.68	85.39	24.47	5.20
SpeedBoat	BiLSTM	88.06	86.22	87.18	86.57	7.74	19.80
KaiYuan	BiLSTM	85.50	82.91	83.79	83.27	24.64	9.54

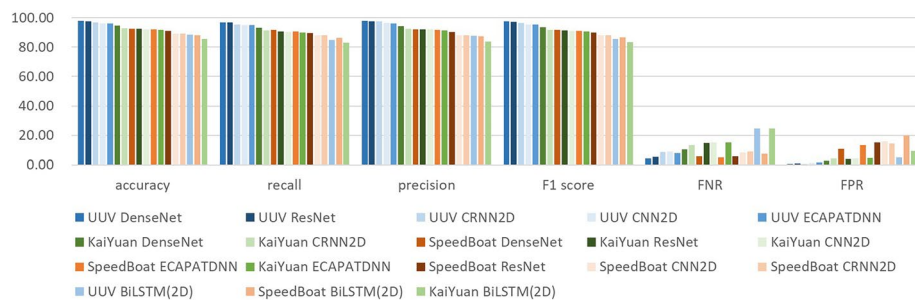


Fig. 11 Average experimental results of deep learning methods for 2D feature classification. Each legend in the figure represents the classification result of a particular network on a specific dataset. The average indicators are calculated with different datasets using the same network. We represent all UUV experiment outcomes with blue bars, all KaiYuan experiment results with green bars, all SpeedBoat experiment outcomes with orange bars. Darker colors in the bars indicate higher computational complexity. The data can be obtained from Table 6

ResNet, CNN, CRNN, and BiLSTM. In the UUV experiment using the DenseNet network, the average recognition accuracy can reach 97.78%, 96.96% of the data containing UUV targets are accurately recognized, and 97.93% of the data predicted by the network containing UUV targets are correct targets. The average F1 score also reaches 97.36%, with only 4.34% of the data containing UUV targets being falsely recognized, and the mispredicted UUV accounts for only 0.84% of the non-target ship samples.

Combining the results from the three experiments, it is evident that the networks demonstrating the best overall performance for 2D feature classification are DenseNet and CRNN, followed by ResNet, which exhibits certain advantages for small-scale datasets (UUV). ECAPA-TDNN, on the other hand, presents a more

pronounced advantage in classifying large-scale and complex datasets (SpeedBoat). CNN performs at an average level, while the network with the poorest classification performance is BiLSTM.

To better assess network performance, we computed the scale and computational complexity of deep learning models, primarily using two indicators: model parameters and FLOPs (floating point operations per second). Model parameters refer to the total number of trainable parameters in a deep learning model, including weights and biases in neural networks. A higher number of parameters indicate a larger model capacity, but it also implies higher computational costs for training and inference. FLOPs represent the number of floating-point operations per second, commonly used to measure the performance of computers or computing devices. In deep learning, FLOPs are often used to describe the computational complexity of a model, indicating the total number of floating-point operations. A higher FLOPs value suggests that the model requires more computational resources. These two metrics provide comprehensive insights into the scale and computational requirements of a model.

The computational experimental results, as presented in Table 7, are visually presented in Fig. 11 by incorporating computational complexity into the classification outcomes. In the figure, darker colors in the bars indicate higher computational complexity. From these two perspectives, it is evident that DenseNet and ResNet exhibit higher computational complexity and, correspondingly, superior classification performance. Models with moderate computational efficiency, such as ECAPA-TDNN and BiLSTM, demonstrate relatively poorer classification results. On the other hand, CNN and CRNN, being lightweight models, achieve above-average classification performance. This further underscores the principle that there is no free lunch, and pursuing exceptional accuracy inevitably requires sacrificing computational efficiency at a certain cost.

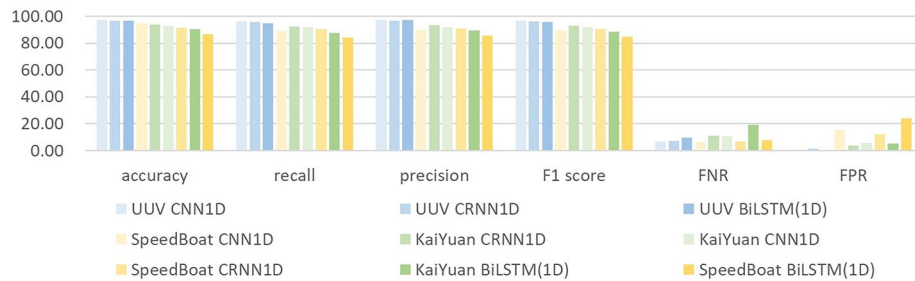
In conclusion, for underwater acoustic target recognition tasks using 2D features, CRNN is recommended due to its lower computational complexity and relatively good classification performance. ECAPA-TDNN is suggested. It has moderate computational complexity but demonstrates a certain advantage in handling large-scale datasets. If the task goal is achieving extremely high classification accuracy without considering computational efficiency, DenseNet can be directly chosen. It consistently presents the best classification results across datasets of different scales. Empirically, similar choices in practical applications have indeed yielded better classification results, consistent with our experimental findings.

Table 7 Computational complexity and scale of deep learning models

Method	FLOPs	Parameters
CNN2D	0.96M	0.33M
CRNN2D	4.76M	0.07M
DenseNet	933.09M	7.97M
ECAPA-TDNN	169.44M	1.53M
BiLSTM	102.24M	0.80M
ResNet	829.26M	13.96M

Table 8 Average results (%) of deep learning methods for 1D feature classification. We averaged the experimental results of all 1D features under the same network and dataset

Experiment	Net	Accuracy	Recall	Precision	F1 score	FNR	FPR
UUV	CNN1D	97.22	96.17	97.36	96.70	6.62	1.04
UUV	CRNN1D	96.80	95.72	96.77	96.18	7.16	1.39
UUV	BiLSTM	96.50	94.84	97.08	95.69	9.59	0.33
SpeedBoat	CNN1D	94.82	89.14	89.87	89.46	6.41	15.32
KaiYuan	CRNN1D	93.82	92.52	93.34	92.87	11.25	3.70
KaiYuan	CNN1D	92.74	91.94	91.84	91.82	10.37	5.74
SpeedBoat	CRNN1D	91.53	90.67	90.72	90.64	6.51	12.16
KaiYuan	BiLSTM	90.24	87.79	89.71	88.58	19.34	5.08
SpeedBoat	BiLSTM	86.54	84.08	85.71	84.59	7.84	24.00

**Fig. 12** Average experimental results of deep learning methods for 1D feature classification. The meaning of the legends, the color schemes and shades, and the data sources are similar to those in Fig. 11

Comparison of network for 1D feature classification

The network classification results for 1D data are illustrated in Table 8 and Fig. 11. In the UUV experiment, the CNN network outperformed CRNN and BiLSTM. In the KaiYuan experiment, CRNN demonstrated superior performance to CNN based on FPR and FNR indicators; however, CNN outperformed CRNN based on the other four metrics. In the SpeedBoat experiment, CNN exhibited better accuracy compared to CRNN, whereas CRNN outperformed CNN in terms of FPR, F1 score, and recall metrics. The performance of both networks was relatively comparable in terms of FNR and precision. It is worth noting that BiLSTM consistently showed poorer performance across all experiments. With the exception of the SpeedBoat experiment based on BiLSTM, the classification results for all other networks consistently achieved above 90%. Notably, the UUV experiment using CNN1D demonstrated an impressive average classification result of 97.22%.

Combining the above experimental results, we can infer that 1D features contain relatively simple information, and satisfactory results can be achieved by employing straightforward network classification. Therefore, for underwater acoustic target recognition tasks based on 1D features, we propose the following strategy: For small-scale datasets (UUV), CNN networks can be used for classification, while for medium- and large-scale datasets (KaiYuan and SpeedBoat), CRNN networks can be employed for classification.

Comparison of network for three datasets experiments

Based on Fig. 11 and 12, it can be observed that in the majority of metrics, the UUV experiment outperforms the KaiYuan experiment, while the results of the KaiYuan experiment are also superior to those of the SpeedBoat experiment. Particularly notable is the FPR metric, where the indicator FPR is significantly higher in the SpeedBoat experiment compared to the UUV and KaiYuan experiments. This is primarily due to the diverse range of vessels included in the SpeedBoat experiment, resulting in more complex combinations of different ships. We have depicted pie charts corresponding to the vessels in each of the three experimental datasets, as shown in Fig. 13. The SpeedBoat experiment comprises a total of 2309 data points, involving 13 types of ships such as QianDao and GreenCity; the KaiYuan experiment includes 649 data points with ships such as WorkShip and PoliceBoat; the UUV experiment, with only 300 data points, features vessels such as PoliceBoat and FishBoat. Under the same feature and network conditions, the model exhibits better classification performance on smaller and less complex datasets, leading to the result where the UUV experiment performs better than the KaiYuan experiment, which, in turn, outperforms the SpeedBoat experiment.

Furthermore, in the evaluation of the FNR metric, we observed that the KaiYuan experiment had a higher FNR than the UUV experiment, and both were higher than the SpeedBoat experiment, contrary to the conclusions drawn earlier. We speculate that the reason of this experimental result is related to the composition of the data. Firstly, the SpeedBoat dataset contains 9% single-target data, which is advantageous for the model to learn the characteristics of SpeedBoat targets, resulting in a low false-negative rate. For the KaiYuan dataset, although it includes 1% single-target data, the proportion is too small. Additionally, compared to the UUV data, there is a serious imbalance in the types of samples, which may be the reason for the slightly higher false-negative rate in the KaiYuan dataset. As for the UUV dataset, although it does not contain single-target data, its sample combinations are relatively simple and evenly distributed, resulting in a moderate false-negative rate.

4.2.2 Comparison of feature

In the feature comparison analysis, we consider that an excellent feature must be applicable in different environments and networks. Therefore, we directly average the results

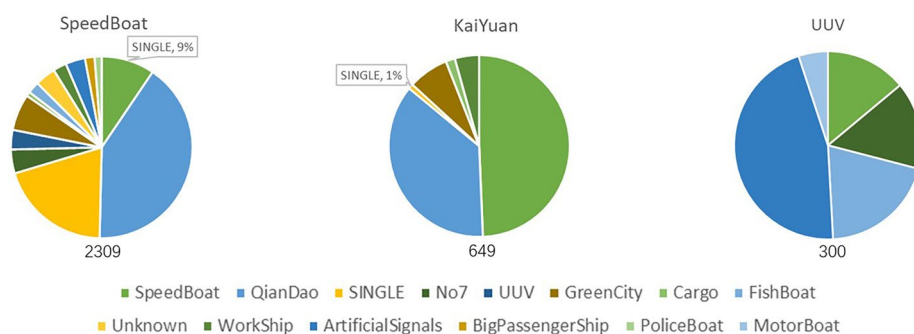


Fig. 13 The data distribution for the three experiments. In the SpeedBoat experiment, the total number of vessels is 2309, while in the KaiYuan experiment, the total number of vessels is 649, and in the UUV experiment, the total number of vessels is 300. Since our datasets contain multiple targets, the pie charts reflect the occurrence frequency of the targets

Table 9 Average results (%) of the same feature

Feature	Accuracy	Recall	Precision	F1 score	FNR	FPR
Spectrum	96.73	96.43	96.30	96.35	3.49	3.65
MFCC	95.47	94.92	95.06	94.95	5.36	4.54
Log Mel	94.24	93.24	93.89	93.48	7.43	6.09
PSD	93.73	92.40	93.55	92.87	7.84	7.35
GFCC	92.77	91.43	92.25	91.75	10.23	6.66
PNCC	92.56	91.25	91.96	91.54	10.16	7.25
DEMON	89.61	85.46	87.62	86.29	17.02	11.92
LOFAR	87.22	83.72	86.89	84.83	20.99	11.58

obtained for each feature, as shown in Table 9. Among all features, the spectrum performs the best while MFCC features perform the best among the 2D features, with both achieving accuracies of over 95%. Our experimental results indicate that the spectrum and MFCC features of ship-radiated noise play crucial roles in classification. The spectrum of ship-radiated noise consists of continuous, line, and modulation spectra, providing essential information about ship operations, engine status, and underwater environments. Meanwhile, MFCC features, designed based on human auditory perception, process signals through Mel filters and employ methods such as logarithmic scaling and discrete cosine transform (DCT) to reduce redundant features. This approach better captures features at different frequencies in ship-radiated noise.

Regarding other features, PNCC utilizes power-law nonlinearity instead of the traditional logarithmic nonlinearity in MFCC coefficients. It introduces a noise suppression algorithm based on asymmetric filtering to mask background excitation and a module for implementing temporal masking. However, these adjustments are disadvantageous for recognizing ship noise signals. GFCC features, employing Gammatone filters, enhance recognition of impulse signals. Still, as ship noise contains very few impulse signals, their recognition effect is not as robust as that of MFCC features. DEMON spectral features respond to information such as the number of propeller blades on a ship, but since the signals identified in this paper are multi-target signals, this feature poorly distinguishes between different signals. LOFAR features are more suitable for finding narrow-band features in signals and low signal-to-noise ratio situations.

4.2.3 Summarize

Based on the raw experimental results, we observe that in the KaiYuan experiment, the best-performing recognition network is the combination of CRNN network with spectrum features. In the SpeedBoat experiment, the best-performing recognition network is the combination of CNN network with spectrum features. In the UUV experiment, the best-performing recognition network is the combination of BiLSTM network with spectrum features. Spectrum features perform excellently in all three experiments, and the complete experimental results for this feature are shown in Table 10. The highest accuracy for features containing targets in the three experiments can reach 97.15%, 95.40%, and 99.56%, respectively. Moreover, they exhibit extreme stability across ten tests, with the vast majority of variances nearly approaching zero. Even with networks like BiLSTM

Table 10 The classification results (%) of spectrum feature

Data	Net	Accuracy	Recall	Precision	f1	FNR	FPR
SpeedBoat	CNN1d	95.40	94.57	95.27	94.89	2.69	8.18
	CRNN1D	94.09	93.71	93.34	93.52	5.03	7.55
	BiLSTM	93.66	93.50	92.69	93.07	5.99	7.00
KaiYuan	CNN1d	97.11	96.71	96.74	96.72	4.44	2.14
	CRNN1D	97.15	97.15	96.47	96.80	2.87	2.84
	BiLSTM	95.45	94.82	94.86	94.84	7.03	3.34
UUV	CNN1d	98.64	98.49	98.38	98.43	1.91	1.11
	CRNN1D	99.49	99.23	99.60	99.41	1.47	0.07
	BiLSTM	99.56	99.68	99.32	99.49	0.00	0.63

that demonstrate comparatively poor overall performance, using spectrum features can still yield high accuracy. This also indirectly indicates that in the field of underwater target recognition, the importance of feature decision-making may surpass that of network decision-making.

Based on the above experimental task results, in underwater target recognition task of identifying a specific type of ship signals from multiple targets, spectrum is preferred as the choice for 1D features, while MFCC features are preferred for 2D features. Secondly, for 1D feature classification, CNN can be chosen for small-scale datasets, while CRNN is suitable for medium- to large-scale datasets. For 2D feature classification, CRNN is recommended due to its lower computational complexity and relatively good classification performance. ECAPA-TDNN is suggested. It has moderate computational complexity but demonstrates a certain advantage in handling large-scale datasets. If the task goal is achieving extremely high classification accuracy without considering computational efficiency, DenseNet can be directly chosen. It consistently presents the best classification results across datasets of different scales. Empirically, similar choices in practical applications have indeed yielded better classification results, consistent with our experimental findings.

5 Conclusion

This paper introduces QiandaoEar22, an underwater acoustic multi-target mixing dataset constructed through the field experiment collection in Qiandao Lake. The dataset contains 9 h and 28 min of ship-radiated acoustic noise and 21 h and 58 min of background noise. To demonstrate the availability of QiandaoEar22, we executed two experimental tasks. The first task focuses on assessing the presence of ship-radiated noise, while the second task involves identifying specific ships within the recognized targets in the multi-ship mixed data. For latter task, we extracted eight features from the data and employed six deep learning networks for classification, aiming to evaluate and compare their performance. We set SpeedBoat, passenger ship KaiYuan, and underwater unmanned vehicle (UUV) as the research object. The best recognition accuracy of UUV target is 97.78%, and we found using spectrum and MFCC as feature inputs and DenseNet as classifier can achieve excellent recognition performance. Considering the computational efficiency, CRNN and ECAPA-TDNN are also good choices. In the future, we hope to establish some form of collaborative network to better learn the

semantic information contained in different features, and to develop more efficient deep learning networks with higher robustness to better accomplish underwater acoustic target recognition tasks.

Acknowledgements

This work was funded jointly by Natural Science Foundation of Shanghai (Grant No. 22ZR1475700), Youth Innovation Promotion Association CAS (Grant No. 2021022), and the development fund for Shanghai talents. We also sincerely thank Prof. Weijie Xu, Prof. Weicai Zhang, and Mr. Bang Jin for their great support during the experiment.

Author Contributions

Xiaoyang Du contributed to the coding, analysis of experimental data, and drafting of the original manuscript. Feng Hong contributed to the acquisition of the research project, design and execution of the experimental data collection scheme, guidance on algorithm analysis research, and provided advices on the revision of the draft manuscript. All authors have read and agreed to the manuscript.

Funding

This work was funded jointly by Natural Science Foundation of Shanghai (Grant No. 22ZR1475700), Youth Innovation Promotion Association CAS (Grant No. 2021022), and Young Talent Cultivation Program of Shanghai Branch of CAS.

Availability of data and materials

The authors confirm that the data supporting the findings of this study are available within the article.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

All authors consent to the publication of this manuscript.

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Received: 12 April 2024 Accepted: 2 September 2024

Published online: 08 November 2024

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